

Java Assignment 02: Control Flow & Static Methods

1. Electricity Bill Calculation — if-else-if

Problem Statement:

Write a Java program to calculate an electricity bill based on the number of units consumed.

Read the number of units in `main()`. Pass it to a static method `calculateBill()`. The method should calculate and return the bill amount to `main()`.

Concepts: `int`, `double`, arithmetic operator, `if-else-if`, static method, return value.

2. Student Grade Calculation — if-else-if

Problem Statement:

Write a Java program to calculate a student's grade based on marks.

Read the marks in `main()`. Pass the marks to a static method `calculateGrade()`. The method should return the grade to `main()`.

Concepts: `int`, relational operators, `if-else-if`, `char/String`, static method.

3. Product Discount — if-else-if

Problem Statement:

Write a Java program to calculate the final price of a product after applying a discount.

Read `productId`, `productName`, and `price` in `main()`. Pass the price to a static method `calculateFinalPrice()`. The method should calculate and return the final price.

Display the product details and final price in `main()`.

Concepts: `int`, `String`, `double`, arithmetic operators, `if-else-if`.

4. Largest of Two Numbers — if-else

Problem Statement:

Write a Java program to find the largest of two numbers.

Read two integers in `main()`. Pass both numbers to a static method `findLargest()`. The method should compare the numbers and return the larger number to `main()`.

If both numbers are equal, return either number.

Concepts: `int`, relational operator, `if-else`, static method, parameters, return.

5. Check Voting Eligibility — if

Problem Statement:

Write a Java program to check whether a person is eligible to vote.

Read the person's age in main(). Pass the age to a static method isEligible(). The method should return true if the age is 18 or above, otherwise return false.

Display an appropriate message in main() based on the returned result.

Concepts: int, relational operator, boolean, if, static method.

6. Employee Salary Calculation — if-else

Problem Statement:

Write a Java program to calculate an employee's final salary.

Read the employee's basic salary in main().

If basic salary is $\geq 50,000$ or above, add 20% HRA.

Otherwise, add 10% HRA.

Pass the basic salary to a static method calculateSalary(). The method should calculate and return the final salary.

Concepts: double, arithmetic operators, if-else, percentage calculation, static method.

7. Temperature Conversion — Operators + if

Problem Statement:

Write a Java program to convert temperature from Celsius to Fahrenheit.

Formula:

$$\text{Fahrenheit} = (\text{Celsius} \times 9 / 5) + 32$$

Read the Celsius temperature in main(). Pass it to a static method convertTemperature(). The method should return the Fahrenheit temperature.

In main(), also check whether the converted temperature is above 100°F and display an appropriate message.

Concepts: double, arithmetic operators, if, static method, return value.

8. Simple Calculator — if-else-if

Problem Statement:

Write a Java program to perform a basic arithmetic operation.

Read two numbers and an operator (+, -, *, /) in main().

Pass all three values to a static method calculate().

The method should perform the requested operation and return the result.

For division, handle division by zero using a conditional statement.

Concepts: double, char, arithmetic operators, if-else-if, parameters, return value.

9. Bus Ticket Fare — if-else-if

Problem Statement:

Write a Java program to calculate a bus ticket fare based on passenger age.

Read the passenger's name and age in main(). Pass the age to a static method calculateFare().

The method should return the ticket fare to main().

Display the passenger name, age, and fare.

Concepts: String, int, if-else-if, relational operators, static method.

10. Mobile Data Usage — if-else-if

Problem Statement:

Write a Java program to calculate the mobile data charge based on data usage.

Read the customer's mobile number and data usage in GB in main().

Pass the data usage to a static method calculateCharge().

The method should calculate and return the charge to main().

Display the mobile number, data usage, and final charge.

Concepts: long, double, arithmetic, if-else-if, static method, parameters, return.

Units | Rate

0–100 | ■2 per unit

101–200 | ■3 per unit

201–300 | ■5 per unit

Above 300 | ■7 per unit

Marks | Grade

90–100 | A

75–89 | B

60–74 | C

50–59 | D

Below 50 | F

Purchase Amount | Discount

■10,000 and above | 20%

■5,000–■9,999 | 10%

■2,000–■4,999 | 5%

Below ■2,000 | No discount

Age | Fare

Below 5 | Free

5–12 | ■20

13–59 | ■40

60 and above | ■25

Data Used | Charge

Up to 1 GB | ■50

More than 1 GB up to 5 GB | ■100

More than 5 GB up to 10 GB | ■200

Above 10 GB | ■350